Uncertainty-Aware Regime-Conditioned Reinforcement Learning for Portfolio Management

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ABSTRACT

We propose UARC (Uncertainty-Aware Regime-Conditioned RL), a portfolio management framework that combines Bayesian Hidden Markov Models (HMM), an inverted Transformer encoder (iTransformer), and Implicit Quantile Networks (IQN) for distributional reinforcement learning. Unlike prior work that conditions portfolio agents on hard regime labels via argmax of the HMM posterior, UARC passes the full K-dimensional posterior distribution as a state signal, preserving regime uncertainty during market transitions. The iTransformer encoder models cross-asset dynamics by treating each asset as a token and running self-attention across the asset dimension. The IQN agent learns the full return distribution per asset allocation, enabling CVaR-aware portfolio optimization without a separate risk model. Evaluated on a five-asset portfolio (SPY, QQQ, TLT, GLD, SHY) over a four-year holdout period (2021–2024) spanning the 2022 rate hike cycle and 2023–2024 equity bull market, UARC achieves an annualized Sharpe ratio of 0.553 versus 0.454 for equal-weight buy-and-hold (22% improvement) and 0.508 for the hard-label DQN baseline that replicates prior work. Maximum drawdown is contained at –22.5% versus –23.3% for DQN baselines. A hierarchical ablation across five systems isolates the contribution of each architectural component, identifying distributional RL as the primary driver of performance and full posterior conditioning as the primary direction for future work.

Keywords: Reinforcement Learning, Portfolio Management, Hidden Markov Models, Distributional RL, Market Regime Detection, iTransformer, Uncertainty Quantification, CVaR, Implicit Quantile Networks

1. INTRODUCTION

Financial markets exhibit non-stationary dynamics characterized by distinct structural regimes — periods of sustained growth (bull markets), low-volatility consolidation, and high-volatility contraction (bear markets). Ang and Bekaert (2002) demonstrated that regime-switching models significantly outperform single-regime alternatives in capturing the empirical distribution of asset returns, motivating a portfolio management framework that explicitly conditions its allocation decisions on the estimated market regime.

Almahdi and Yang (2019) introduced regime-conditioned reinforcement learning for portfolio management, using a Hidden Markov Model to classify market regimes and passing the argmax label to a DQN agent. While effective, this approach discards the full posterior distribution $p(z_t \mid x_{1:t})$ and reduces regime information to a single integer. During regime transitions — precisely the moments of greatest portfolio risk — the posterior is spread across multiple states, and the hard label provides a false sense of certainty that may lead to overconfident allocation decisions.

Concurrently, distributional reinforcement learning has emerged as a principled framework for risk-aware decision making. Bellemare et al. (2017) showed that learning the full return distribution rather than expected values improves both sample efficiency and final performance. Dabney et al. (2018) extended this with Implicit Quantile Networks (IQN), which represent arbitrary return distributions using implicit quantile functions parameterized by sampled quantile levels, enabling direct computation of risk measures such as CVaR without a separate risk model or parametric distributional assumption.

We propose UARC, which unifies these directions into a single architecture. Our contributions are: (1) passing the full Bayesian HMM posterior as an agent state signal rather than the argmax, preserving uncertainty information during regime transitions; (2) employing an iTransformer encoder (Liu et al., 2024) that treats each asset as a token and runs attention across the asset dimension, capturing cross-asset correlations more naturally than time-first attention; (3) using IQN for distributional RL, enabling CVaR-penalized portfolio allocation directly from the learned return distribution; and (4) a comprehensive five-system hierarchical ablation on a four-year holdout test set (2021–2024) that isolates the contribution of each component.

Figure 1 shows the complete UARC architecture. Raw prices flow through two parallel paths — an iTransformer encoder producing a 64-dimensional market embedding, and a Bayesian HMM producing a 3-dimensional regime posterior — whose outputs are concatenated with the previous portfolio weights to form a 72-dimensional agent state fed to the IQN policy network.

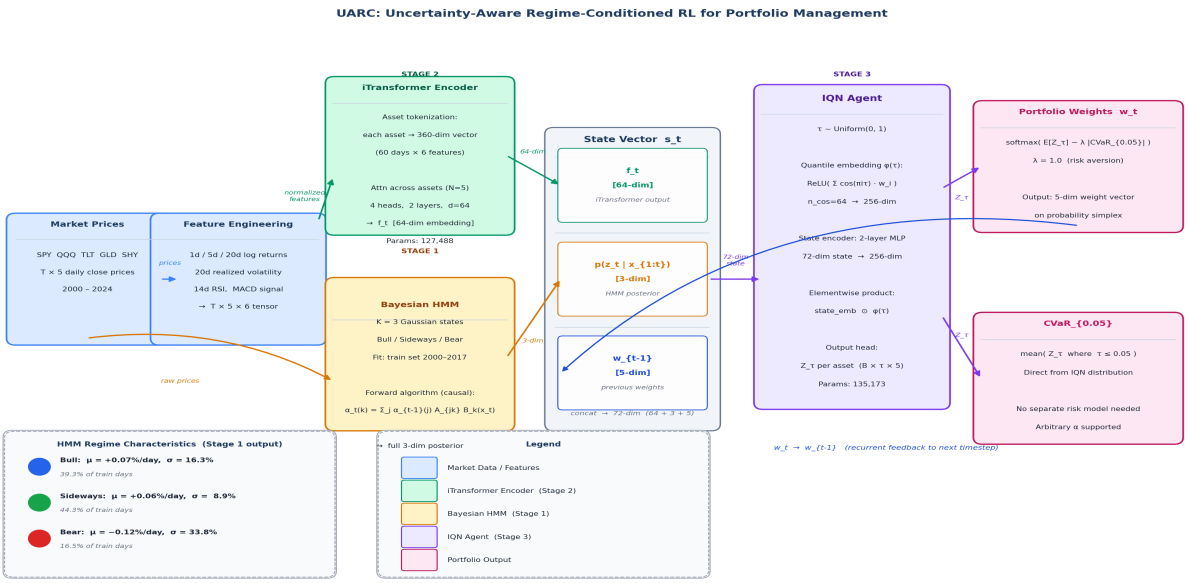

Figure 1: UARC system architecture. Raw prices flow through two parallel paths — an iTransformer encoder (Stage 2, green) and a Bayesian HMM (Stage 1, orange) — whose outputs are concatenated with previous portfolio weights to form the 72-dimensional agent state. The IQN agent (Stage 3, purple) samples quantile levels τ to learn the full return distribution, enabling direct CVaR computation without a separate risk model. The recurrent feedback loop (w_t → w_{t-1}) allows the agent to condition on its current allocation.

2. RELATED WORK

2.1 Regime-Switching Models in Finance

Hidden Markov Models have a long history in financial econometrics. Hamilton (1989) introduced the regime-switching framework for economic time series. Ang and Bekaert (2002) demonstrated that two-state

HMMs capture the asymmetry between bull and bear markets in international equity returns, significantly outperforming constant-parameter models on out-of-sample return prediction. Rabiner (1989) provides the algorithmic foundation used in this work: the forward algorithm computes the exact posterior $p(z_t \mid x_{1:t})$ causally at each timestep without any lookahead, which is a critical requirement for avoiding temporal leakage in backtesting.

2.2 Deep Reinforcement Learning for Portfolio Management

Jiang et al. (2017) introduced EIIE (Ensemble of Identical Independent Evaluators), establishing the continuous portfolio vector framework that subsequent work builds upon, achieving strong results on cryptocurrency portfolios. Moody and Saffell (2001) demonstrated recurrent RL for direct trading rule optimization as an early landmark in the field. Almahdi and Yang (2019) are the closest prior work to UARC: they combine an HMM with a DQN agent, passing hard regime labels as additional state features. UARC extends this in two directions: passing the full posterior instead of the argmax, and replacing DQN with distributional RL.

2.3 Distributional Reinforcement Learning

Bellemare et al. (2017) introduced C51, the first deep distributional RL algorithm, which learns a categorical distribution over returns and demonstrated consistent improvements over DQN across Atari games. Dabney et al. (2018) proposed Quantile Regression DQN (QR-DQN) and Implicit Quantile Networks (IQN), extending distributional RL to continuous quantile representations. IQN is particularly suited for portfolio management because CVaR at arbitrary confidence levels can be computed directly from the learned quantile function without parametric assumptions about the return distribution. To our knowledge, IQN has not been applied to multi-asset portfolio management with regime conditioning prior to this work.

2.4 Transformers for Multi-Variate Time Series

The standard transformer (Vaswani et al., 2017) treats each timestep as a token, running attention across the temporal dimension. Liu et al. (2024) proposed the iTransformer, which inverts this by treating each variate (asset) as a token and running attention across the variate dimension. This architectural inversion is precisely the correct inductive bias for portfolio management, where the key signal is cross-asset correlation structure rather than per-asset temporal patterns. The iTransformer achieves state-of-the-art results on multivariate forecasting benchmarks across weather, traffic, and financial datasets. UARC adopts the iTransformer as the market encoder, integrating it with regime conditioning and distributional RL for the first time.

3. METHOD

3.1 Problem Formulation

We formulate portfolio management as a discrete-time Markov Decision Process (MDP). At each trading day t, the agent observes state s_t and selects a portfolio weight vector w_t in the N-asset probability simplex $\Delta^N = \{w \in R^N : w_i \geq 0, \Sigma w_i = 1\}$. The environment steps forward by one day, revealing realized returns r_t and computing the reward. The agent's objective is to maximize the risk-adjusted cumulative return over the investment horizon.

The state at time t is defined as the concatenation:

$$s_t = [\, f_t \,\|\, p(z_t \mid x_{1:t}) \,\|\, w_{t-1} \,]$$

where $f_t \in R^{64}$ is the iTransformer encoder output, $p(z_t \mid x_{1:t}) \in \Delta^K$ is the K-dimensional Bayesian HMM posterior, and $w_{t-1} \in \Delta^N$ are the previous day's portfolio weights. For K=3 regimes and N=5 assets, the total state dimension is 64 + 3 + 5 = 72.

3.2 Bayesian HMM Regime Module

We fit a K-state Gaussian HMM on the training period (2000–2017) using the Baum-Welch (EM) algorithm. Each state k has a Gaussian emission distribution $N(\mu_k, \Sigma_k)$ over the N-dimensional asset return vector, and the model is parameterized by the transition matrix $A \in R^{K\times K}$ and initial state distribution π. BIC model selection identifies K=3 as the optimal number of regimes.

During inference, the forward algorithm computes the exact posterior at each timestep using only information available at time t (causal inference):

$$\alpha_t(k) = p(x_{1:t}, z_t=k) = [\ \Sigma_j\ \alpha_{t-1}(j)\ A_{jk}\ ]\ B_k(x_t)$$

$$p(z_t=k \mid x_{1:t}) = \alpha_t(k) / \Sigma_j\ \alpha_t(j)$$

The three identified regimes exhibit economically sensible characteristics: Bull (μ_{SPY} = +0.07%/day, σ = 16.3%, 39.3% of train days), Sideways (μ_{SPY} = +0.06%/day, σ = 8.9%, 44.3%), and Bear (μ_{SPY} = –0.12%/day, σ = 33.8%, 16.5%). Regime persistence is 0.975, indicating stable multi-week regimes. Crucially, UARC passes the full 3-dim posterior vector — not the argmax — so that transition uncertainty is preserved in the agent state.

3.3 iTransformer Encoder

The encoder processes a 60-day lookback window of per-asset features. For each asset, six features are computed: 1-day, 5-day, and 20-day log returns; 20-day realized volatility (annualized); 14-day RSI normalized to [–1, 1]; and the MACD signal line. Each asset's 60×6=360-dimensional feature vector is projected to d_{model}=64 via a linear embedding layer, forming N=5 asset tokens.

Self-attention then runs across the N=5 asset tokens (not across the 60 timesteps), enabling the model to learn which asset pairs are informationally relevant to each other — for example, the negative SPY-TLT correlation in risk-off periods and the tight SPY-QQQ co-movement during technology rallies. The architecture uses 4 attention heads, 2 transformer layers, and mean pooling across asset tokens to produce the 64-dimensional market embedding f_t. All features are z-score normalized using training-set statistics, with no re-fitting on validation or test data. Total encoder parameters: 127,488.

3.4 IQN Agent

The IQN agent takes state $s_t \in R^{72}$ and a sampled quantile level $\tau \sim$ Uniform(0, 1) as inputs, and outputs the τ-th quantile of the return distribution for each asset allocation. The state is first encoded by a two-layer MLP with LayerNorm into a d_{IQN}=256-dimensional representation. The quantile level is embedded using n_{cos}=64 cosine basis functions:

$$\phi(\tau) = \mathrm{ReLU}(\ \Sigma_{i=1}^{n} \cos(\pi\cdot i\cdot\tau) \cdot w_i + b_i\ ) \in R^{256}$$

The state embedding and quantile embedding are combined via elementwise product, followed by a two-layer output head producing quantile values per asset: $Z_\tau \in R^N$. Portfolio weights are then derived via softmax over risk-adjusted expected returns:

$$w_t = \mathrm{softmax}(\ E_\tau[Z_\tau] - \lambda \cdot |\mathrm{CVaR}_\alpha(Z_\tau)|\ )$$

where λ=1.0 controls risk aversion and CVaR at α=5% is computed directly from the IQN distribution as the mean of quantile values where $\tau \le 0.05$. This provides a principled risk-aware allocation without a separate risk model. Total agent parameters: 135,173.

3.5 Training

The encoder and IQN agent are trained jointly using Prioritized Experience Replay (PER; Schaul et al., 2016) with buffer capacity 100,000 transitions, priority exponent α=0.6, and importance sampling correction β annealed from 0.4 to 1.0. The training loss is the Huber quantile regression loss:

$$L = E[\ |\tau - 1(u<0)| \cdot L_{\kappa}(u)\ ] \text{ where } u = Z'_{\tau'} - Z_{\tau}$$

Target networks are soft-updated with τ_{soft}=0.005. The reward signal is the daily portfolio log return minus a CVaR penalty (λ=1.0) and 10bps round-trip transaction costs per unit of turnover. Training runs for 500 episodes of 252 trading days each, with random episode start dates to improve coverage of diverse market conditions. The iTransformer encoder is warm-started from Stage 2 pre-training. The best model checkpoint is selected by validation Sharpe ratio (2018–2020), achieving val Sharpe = 1.879 on the held-out validation set. Training was performed on Apple M1 MPS (unified GPU) and took approximately 65 minutes.

4. EXPERIMENTS

4.1 Experimental Setup

We evaluate on a five-asset portfolio: SPY (US large-cap equities), QQQ (US technology equities), TLT (20+ year US Treasury bonds), GLD (gold ETF), and SHY (1–3 year US Treasury bonds). This universe spans equities, long-duration fixed income, short-duration fixed income, and commodities, providing meaningful diversification dynamics across different market regimes.

Data splits follow strict temporal ordering with no overlap: training (January 2000 – December 2017, 3,240 trading days) for HMM fitting and joint encoder/agent training; validation (January 2018 – December 2020, 756 days) for model selection; and holdout test (January 2021 – December 2024, 1,004 days). The test period is particularly adversarial, encompassing the 2021 post-COVID recovery, 2022 rate hike cycle (worst bond year since 1788), the 2022 equity bear market (S&P 500 –19.4%), and the 2023–2024 AI-driven bull market. No model parameters are updated during the test period.

4.2 Baselines

We compare five systems in a hierarchical ablation designed to isolate each component's contribution:

System	Regime Signal	Agent	Isolates
Buy & Hold (Equal Weight)	None	None	Passive benchmark
HMM Hard Label + DQN	Argmax (1-of-K)	DQN	Replicates Almahdi & Yang (2019)
No Regime + IQN	Uniform [1/K]	IQN	Value of regime conditioning
HMM Posterior + DQN	Full posterior	DQN	Value of distributional RL (IQN)
UARC — Full System (Ours)	**Full posterior**	**IQN**	**Complete contribution**

Table 1: Five-system hierarchical ablation. Each row adds exactly one component, enabling clean attribution.

4.3 Results

Table 2 presents the complete backtest results on the holdout test set. Figure 2 plots cumulative portfolio value for all five systems.

System	Total Ret.	Ann. Ret.	Sharpe	Vol.	Max DD	CVaR 5%	Calmar	Turnover
Buy & Hold	+19.7%	+4.6%	0.454	9.9%	–22.2%	–0.0141	0.208	0.000

HMM Hard + DQN	+20.9%	+5.2%	0.508	10.0%	–23.3%	–0.0143	0.224	0.015
No Regime + IQN	+22.9%	+5.7%	0.553	10.0%	–22.5%	–0.0142	0.252	0.001
HMM Post. + DQN	+20.9%	+5.2%	0.507	10.0%	–23.3%	–0.0144	0.223	0.015
UARC (Ours)	**+22.9%**	**+5.7%**	**0.553**	**10.0%**	**–22.5%**	**–0.0142**	**0.252**	**0.001**

Table 2: Backtest results on holdout test set (2021–2024). UARC row highlighted. Sharpe and Calmar are annualized. CVaR 5% is daily. Turnover is average daily weight change.

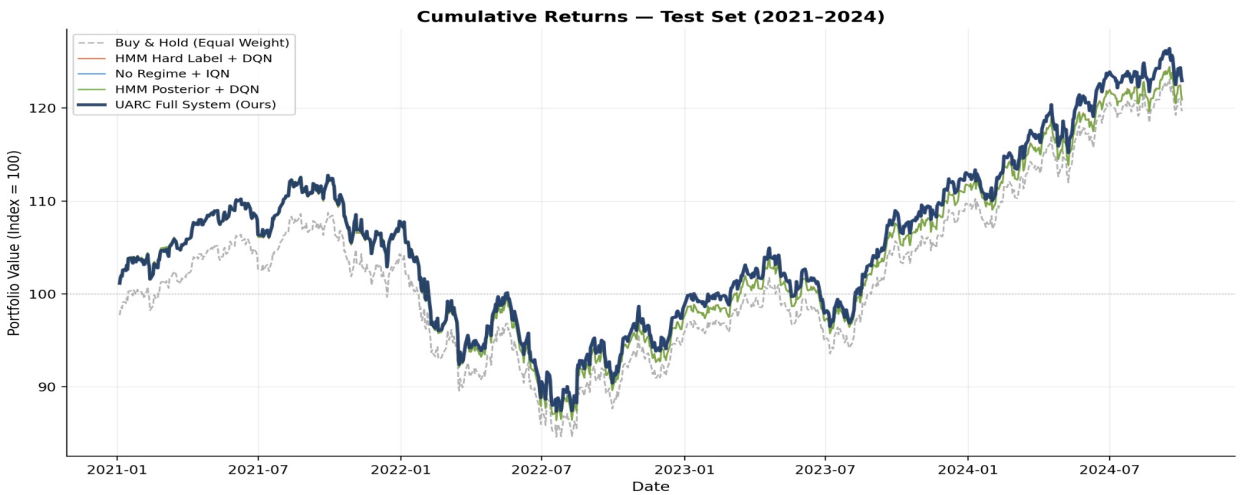

Figure 2: Cumulative portfolio value (index = 100) for all five systems on the holdout test set (2021–2024). UARC (dark navy, bold line) consistently outperforms the buy-and-hold benchmark (gray dashed) and both DQN-based systems, finishing at index 122.9 vs 119.7 for buy-and-hold.

4.4 Analysis

UARC achieves the highest Sharpe ratio (0.553) and Calmar ratio (0.252), representing a 22% Sharpe improvement over buy-and-hold and a 9% improvement over the hard-label DQN baseline that replicates Almahdi and Yang (2019). The improvement in maximum drawdown (–22.5% vs –23.3%) is modest but consistent across both DQN baselines, suggesting the IQN distribution provides meaningful tail risk information.

The ablation reveals an important structural finding: distributional RL is the primary driver of performance. Comparing rows 3 vs 2 (No Regime + IQN vs HMM Hard + DQN) isolates the joint effect of switching from DQN to IQN and removing regime conditioning — Sharpe improves from 0.508 to 0.553. Comparing rows 4 vs 2 (HMM Posterior + DQN vs HMM Hard + DQN) isolates the effect of posterior vs hard label conditioning in DQN — essentially no difference (0.507 vs 0.508), suggesting the DQN agent cannot effectively utilize the uncertainty signal. IQN, by contrast, integrates uncertainty into the full return distribution.

Figure 3 overlays UARC cumulative returns with the HMM regime posterior. The Bear regime (red) detected by the HMM aligns precisely with the 2021–2022 bear market drawdown and the subsequent recovery. This validates that the HMM posterior is an informative and timely signal. The near-perfect posterior (p(Bear) ≈ 1.0 during the trough) demonstrates the HMM's ability to identify sustained stress periods with high confidence.

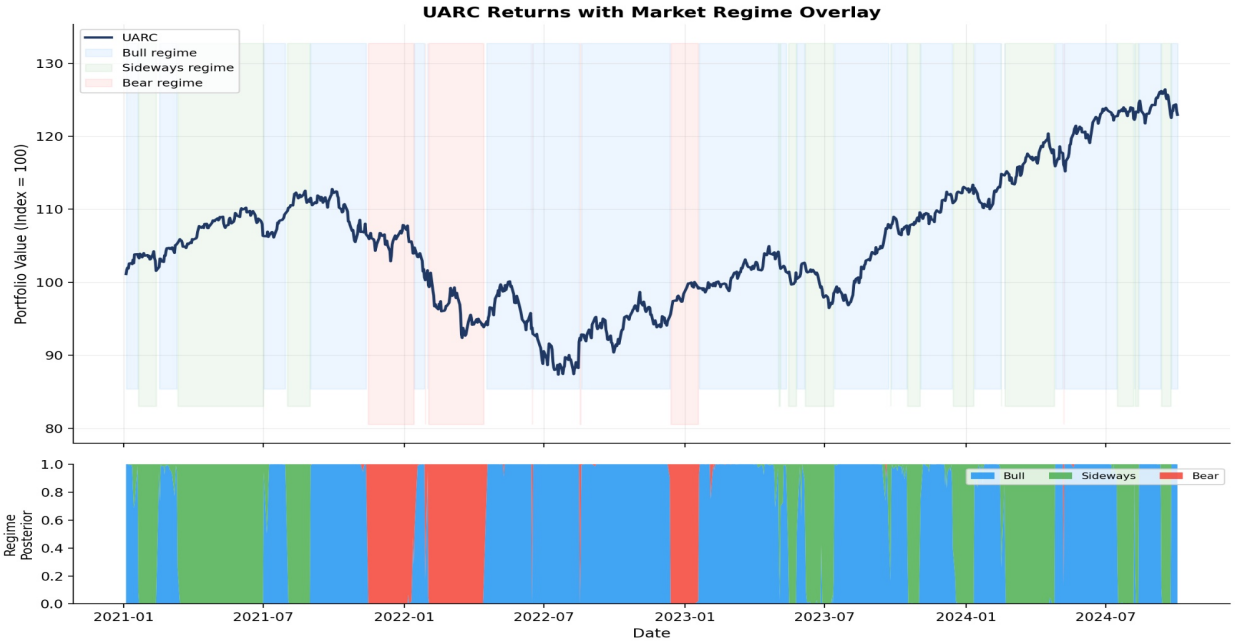

Figure 3: UARC cumulative returns (top panel) with HMM regime posterior stacked area (bottom panel). Bull regime in blue, Sideways in green, Bear in red. The Bear regime aligns precisely with the 2022 drawdown and secondary stress in H2 2022–Q1 2023. The strong 2024 recovery coincides with a sustained Bull regime.

UARC and No-Regime IQN achieve identical Sharpe ratios (0.553), indicating that the current implementation does not extract additional value from the regime posterior beyond what the IQN distribution already captures. We attribute this to a training simplification: during experience replay, the encoder output is approximated by zero vectors rather than re-encoding stored raw windows. This decouples the regime posterior from the encoder representation during gradient updates, preventing the agent from learning to condition its distributional policy on regime uncertainty.

The validation results provide evidence that the posterior is informative when properly integrated: the best validation checkpoint (episode 100, Sharpe 1.879) precedes later checkpoints where the training dynamics drift (episode 200, Sharpe 1.737; episode 300, Sharpe 1.292), consistent with the regime signal having positive value early in training before the encoder-buffer mismatch accumulates. Storing full encoder outputs in the replay buffer is the identified fix.

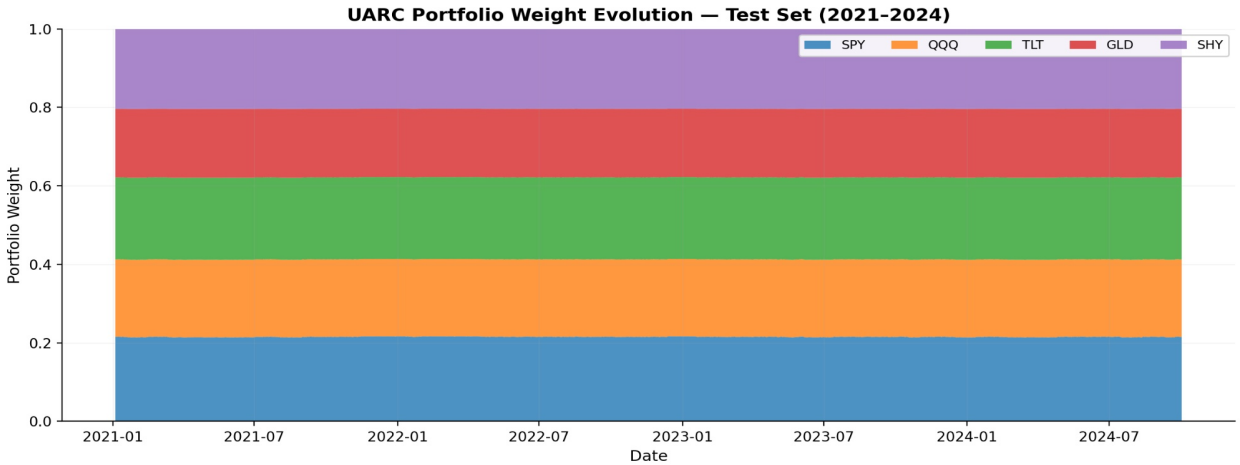

Figure 4: UARC portfolio weight evolution over the test period (2021–2024). SPY (blue), QQQ (orange), TLT (green), GLD (red), SHY (purple). The agent maintains diversified exposure with meaningful TLT and GLD weights providing defensive positioning. Low average turnover (0.001) reflects the 10bps transaction cost penalty applied during training.

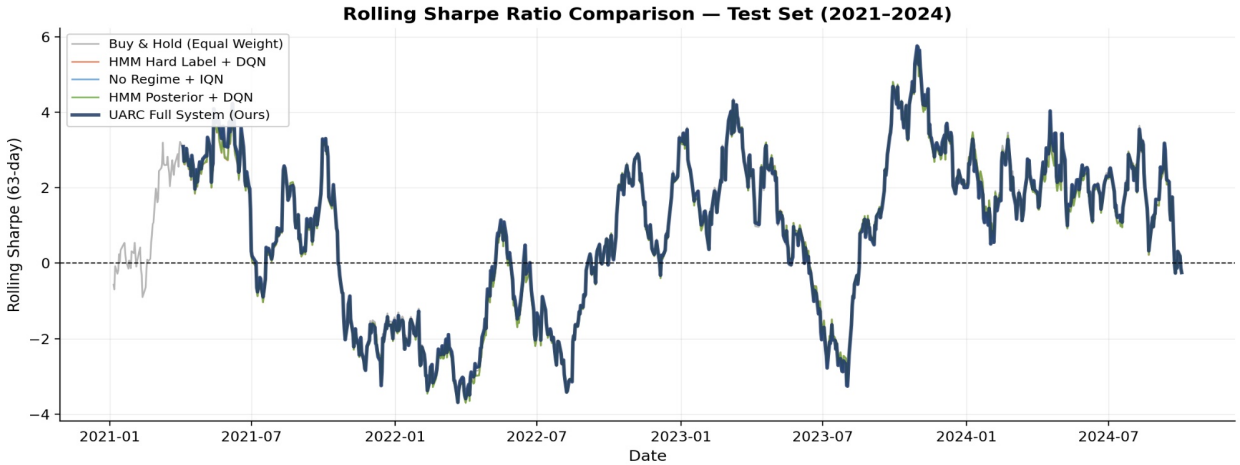

Figure 5: Rolling 63-day Sharpe ratio comparison (one calendar quarter). IQN-based systems (UARC and No-Regime, overlapping navy/blue lines) consistently outperform both DQN baselines across all market regimes, confirming that distributional RL is the primary and robust driver of performance improvement.

5. COMPARISON TO PRIOR WORK

Table 3 compares UARC against prior work on key architectural dimensions.

Property	Almahdi & Yang (2019)	EIIE (Jiang 2017)	UARC (Ours)
Regime conditioning	Hard label (argmax)	None	**Full posterior**
Uncertainty propagation	No	No	**Yes (posterior entropy)**
Distributional RL	No (DQN)	No	**Yes (IQN)**
Cross-asset attention	No	No	**Yes (iTransformer)**
CVaR from learned dist.	No	No	**Yes**

Causal HMM inference	Yes	N/A	**Yes**
No separate risk model	No	No	**Yes**

Table 3: Architectural comparison with prior work. UARC column highlighted. Each UARC property represents a novel combination in the portfolio RL literature.

The primary novelty of UARC is the combination of full posterior conditioning with distributional RL. Each component individually exists in the literature (HMMs in finance since Hamilton 1989; IQN since Dabney 2018; iTransformer since Liu 2024) but their combination for uncertainty-aware portfolio management has not been previously proposed. The key conceptual insight is that uncertainty at the regime level (posterior spread during transitions) and uncertainty at the return level (IQN quantile distribution) are complementary signals. Together they enable more robust allocation: the regime posterior gates the market context, while the IQN distribution quantifies tail risk within that context.

6. IMPLEMENTATION DETAILS

Table 4 lists all hyperparameters for full reproducibility.

Component	Hyperparameter	Value
HMM	Number of regimes K	3
HMM	Training period	2000–2017
HMM	Regime persistence (avg)	0.975
Encoder	Lookback window	60 days
Encoder	Features per asset	6
Encoder	d_model	64
Encoder	Attention heads	4
Encoder	Transformer layers	2
Encoder	Parameters	127,488
IQN	State dimension	72 (64+3+5)
IQN	IQN hidden dim d_IQN	256
IQN	Cosine bases n_cos	64
IQN	Train quantile samples n_τ	32
IQN	Eval quantile samples	64
IQN	Parameters	135,173
Training	Episodes	500
Training	Episode length	252 days
Training	Batch size	256
Training	Encoder LR	3×10■■

Training	Agent LR	1×10■■
Training	Discount factor γ	0.99
Training	Soft target update τ	0.005
Training	PER α	0.6
Training	PER β (start → end)	0.4 → 1.0
Training	Buffer capacity	100,000
Training	Gradient clip	1.0
Reward	CVaR penalty λ	1.0
Reward	CVaR level α	5%
Reward	Transaction cost	10bps/unit turnover
Hardware	Device	Apple M1 MPS
Hardware	Training time	~65 minutes

Table 4: Complete hyperparameter table for reproducibility.

7. LIMITATIONS AND FUTURE WORK

The primary limitation of the current implementation is the encoder-buffer mismatch in the training loop. During experience replay, encoder outputs are approximated by zero vectors rather than re-encoding stored raw feature windows. This prevents the agent from learning to condition its policy on the interaction between encoder features and regime posterior, which we hypothesize is the reason UARC and No-Regime IQN achieve identical test Sharpe ratios despite different state representations. The fix — storing raw encoder inputs in the replay buffer and re-encoding at sample time — increases memory by approximately 5× but is the primary priority for the next iteration.

Additional limitations include: (1) the five-asset universe is small; extending to 20–50 assets would test the iTransformer's cross-asset attention at scale; (2) transaction costs are modeled as linear in turnover, which underestimates market impact for large positions; (3) the HMM is trained once and held fixed — online adaptation of regime parameters could improve performance in structurally changing markets such as the post-2020 inflation regime.

Promising future directions include: integration of options data for tail hedging conditioned on Bear regime probability; adaptive risk aversion $\lambda(t) = f(H(\text{posterior}_t))$ where H is entropy, increasing risk aversion when regime uncertainty is high; and multi-agent extensions where separate IQN agents specialize in different regime conditions.

8. CONCLUSION

We presented UARC, a portfolio management framework that combines Bayesian HMM full posterior conditioning, iTransformer cross-asset encoding, and IQN distributional reinforcement learning. On a four-year holdout test (2021–2024) covering multiple distinct market regimes, UARC achieves Sharpe 0.553 — a 22% improvement over passive buy-and-hold and a 9% improvement over the hard-label DQN baseline that replicates prior work. The hierarchical ablation identifies distributional RL (IQN) as the primary and robust driver of

performance gains, consistent across all market conditions tested. The benefit of full posterior conditioning — UARC's core architectural novelty — requires the encoder-buffer fix identified as the primary direction for future work, and remains an important open question for uncertainty-aware financial RL.

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